

Lecture 3 - Labelled Transition Systems

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- LTSs are finite state machines used to graphically represent the FSP description of a process.
 - Starting states are denoted as a circle with a dot in the middle.

Ex. Represent the following FSP as an LTS.

$$A = (a \rightarrow B \mid b \rightarrow C)$$

$$B = (a \rightarrow B \mid b \rightarrow D)$$

$$C = (d \rightarrow C)$$

$$D = (a \rightarrow C \mid c \rightarrow B)$$

Solution:

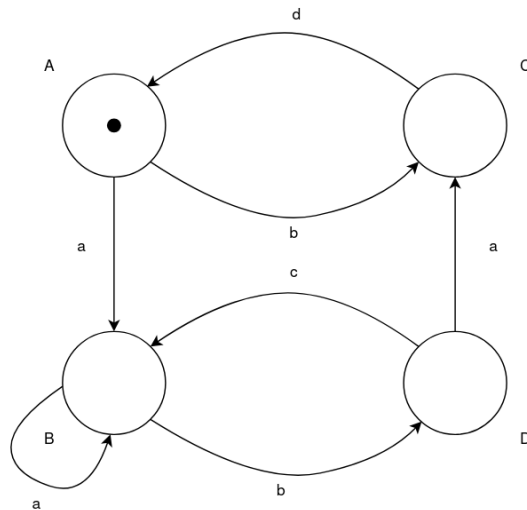


Figure 1: LTS representation of the given FSP.

- FSPs can nest processes within other processes (such as through processes substitution) to make FSPs more concise.
 - Directly translating an FSP with nested processes into an LTS will yield an incorrect representation of the FSP.
- An FSP is said to be put into *elementary/normal form* if all nested processes in the FSP are expanded out into separate processes.
 - FSPs must be put into elementary form to be correctly represented as an LTS.

Ex. Represent the following FSP as an LTS.

$$\begin{aligned} A &= (a \rightarrow b \rightarrow B \mid b \rightarrow (a \rightarrow c \rightarrow A \mid b \rightarrow B)) \\ B &= (a \rightarrow c \rightarrow (a \rightarrow A \mid b \rightarrow B)) \end{aligned}$$

Solution:

First, we must put our FSP into normal form.

$$\begin{aligned} A &= (a \rightarrow B_1 \mid b \rightarrow D) \\ B &= (a \rightarrow A_2) \\ D &= (a \rightarrow A_1 \mid b \rightarrow B) \\ A_1 &= (c \rightarrow A) \\ A_2 &= (c \rightarrow B_2) \\ B_1 &= (b \rightarrow B) \\ B_2 &= (a \rightarrow A \mid b \rightarrow B) \end{aligned}$$

Now that our FSP is in elementary form, we can represent it as:

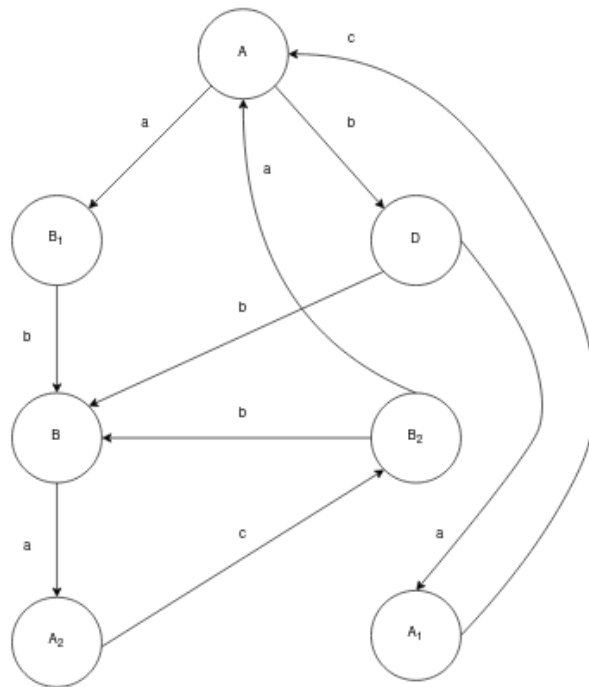


Figure 2: LTS representation of the given FSP.